

LECTURE 22

Multi-Valued Dependencies and Fourth Normal Form

CSC270 · Database Systems

COMSATS University Islamabad, Lahore Campus

What We'll Cover

- Why BCNF alone doesn't catch every source of redundancy
- Multi-valued dependencies: notation $X \twoheadrightarrow Y$ and the intuition behind it
- Trivial vs. non-trivial MVDs, and properties of MVDs
- Inference rules for MVDs, parallel to Armstrong's Axioms
- Fourth Normal Form: definition and test
- Worked example: Employee, Skills, and Languages
- 4NF decomposition and the lossless-join property

Where BCNF Falls Short

`EmpSkillsLangs (empNo, skill, language)`

empNo	skill	language
E1	Java	English
E1	Java	Urdu
E1	Python	English
E1	Python	Urdu
E2	SQL	English

- No non-trivial FD exists: empNo doesn't determine a single skill (E1 has two)
- Only candidate key is the full set — an all-key relation
- So this relation is already in BCNF...
- ...yet E1's 2 skills × 2 languages cross-multiply into 4 rows

BCNF has no mechanism to see this — it only looks at functional dependencies, and there isn't one here.

The Spurious Combination, Made Concrete

Row 2 claims “E1's Java skill is specifically paired with Urdu.” Row 3 claims “E1's Python skill is specifically paired with English.”

Neither claim means anything — skill and language were never actually linked for this employee; they're just two unrelated attributes forced into the same row.

- If E1 learns French, correctness demands inserting TWO new rows (Java, French) and (Python, French)
- Forgetting either one leaves “does E1 speak French” with a different answer depending which row you check
- This is precisely the gap Fourth Normal Form exists to close

Multi-Valued Dependencies

 $X \twoheadrightarrow Y$

“X multi-determines Y” — for every value of X, the set of Y-values is completely independent of the Z-values in that row.

- In EmpSkillsLangs: empNo $\twoheadrightarrow$ skill — for a fixed empNo, the associated skill values are independent of language
- By the same argument, empNo $\twoheadrightarrow$ language holds too — not a coincidence, it's the complementation rule

The intuitive test

For a fixed employee, does knowing which language they speak change which skills are possible? If the honest answer is “no, the two lists are just independently attached” — that's a multi-valued dependency.

Trivial MVDs, and Properties of MVDs

Trivial vs. Non-Trivial

- Trivial: $Y \subseteq X$, or $X \cup Y = R$ (all attributes) — holds automatically, carries no design information
- Non-trivial: neither condition holds — only non-trivial MVDs can violate 4NF
- In EmpSkillsLangs, $\text{empNo} \twoheadrightarrow \text{skill}$ is non-trivial: $\text{skill} \not\subseteq \text{empNo}$, and $\{\text{empNo}, \text{skill}\}$ doesn't cover language

Properties of MVDs

- Every FD is an MVD (not vice versa) — fixed X gives exactly one Y , trivially independent of Z
- MVDs always arrive in a complementary pair: $X \twoheadrightarrow Y$ over $\{X, Y, Z\}$ iff $X \twoheadrightarrow Z$
- Checking an MVD requires looking at the whole relation — not just candidate keys

Deriving New MVDs

Rule	Statement
Complementation	If $X \twoheadrightarrow Y$ holds over R , then $X \twoheadrightarrow (R - X - Y)$ also holds.
Augmentation	If $X \twoheadrightarrow Y$ holds and $W \supseteq Z$, then $WX \twoheadrightarrow YZ$ holds.
Transitivity	If $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$ both hold, then $X \twoheadrightarrow (Z - Y)$ holds.
Replication (FD $\rightarrow$ MVD)	If $X \rightarrow Y$ holds, then $X \twoheadrightarrow Y$ holds.
Coalescence	If $X \twoheadrightarrow Y$ holds, $W \cap Y = \emptyset$, and $W \rightarrow Z$ for some $Z \subseteq Y$, then $X \rightarrow Z$ also holds.

For this course: recognizing complementation on sight, and knowing every FD is automatically an MVD, are the two facts you'll use most.

Fourth Normal Form (4NF)

R is in 4NF if, for every non-trivial MVD $X \twoheadrightarrow Y$ over R, X is a superkey of R.

- A direct, deliberate parallel to BCNF — replace “functional dependency” with “multi-valued dependency”
- Since every FD is also an MVD, 4NF implies BCNF — checking 4NF automatically re-checks every BCNF condition too

4NF's relationship to BCNF

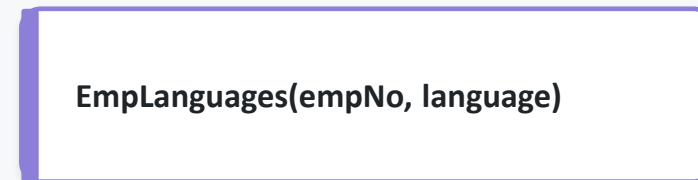
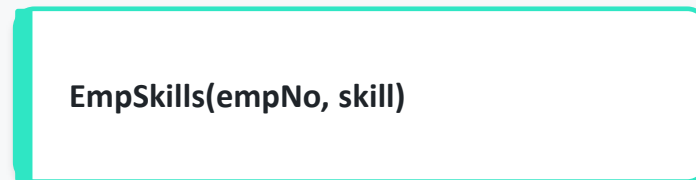
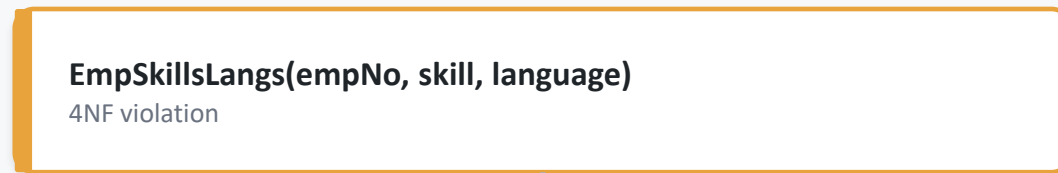
EmpSkillsLangs is in BCNF (no FD violation) but NOT in 4NF — empNo is not a superkey, yet empNo $\twoheadrightarrow$ skill is a genuine non-trivial MVD. 4NF is strictly the next, stronger step after BCNF.

Testing EmpSkillsLangs Against 4NF

- Candidate key: {empNo, skill, language} — the full attribute set (no smaller combination determines the rest functionally)
- Non-trivial MVD $\text{empNo} \twoheadrightarrow \text{skill}$ has determinant {empNo} — a single attribute out of three
- empNo is nowhere close to a superkey — it doesn't even functionally determine skill (E1 has two skill values)

EmpSkillsLangs violates 4NF.

4NF Split of EmpSkillsLangs on empNo → skill



EmpSkills (empNo, skill)

empNo	skill
E1	Java
E1	Python
E2	SQL

EmpLanguages (empNo, language)

empNo	language
E1	English
E1	Urdu
E2	English

Redundancy Removed, Nothing Lost

- Original: 4 spurious cross-multiplied rows for E1 alone
- Decomposed: 5 rows total across both tables, zero spurious pairing
- Adding a third skill for E1 now costs exactly ONE new row in EmpSkills, not two

EmpSkills(empNo, skill)

- Only candidate key: {empNo, skill} — an all-key relation
- Any MVD over a 2-attribute relation is automatically trivial
- 4NF — trivially satisfied

EmpLanguages(empNo, language)

- Identical argument applies
- Only candidate key: {empNo, language} — all-key
- 4NF — trivially satisfied

Lossless-Join Property, Confirmed

Natural join reconstructs the cross-product per employee

`EmpSkills ⋈ EmpLanguages (on empNo)`

empNo	skill	language
E1	Java	English
E1	Java	Urdu
E1	Python	English
E1	Python	Urdu
E2	SQL	English

Fagin's Theorem

R decomposes losslessly into $R_1 = XY$ and $R_2 = XZ$ if and only if $X \twoheadrightarrow Y$ holds over R — exactly the condition we split on.

4NF doesn't reduce information, it removes a physical redundancy — the join still represents every combination; what's eliminated is storing $|skills| \times |languages|$ rows per employee.

Key Takeaways

- A multi-valued dependency $X \twoheadrightarrow Y$ says that for a fixed X , the associated Y -values are completely independent of every other attribute — broader than a functional dependency.
- MVDs always arrive in a complementary pair: $X \twoheadrightarrow Y$ over $\{X,Y,Z\}$ if and only if $X \twoheadrightarrow Z$.
- 4NF requires every non-trivial MVD's determinant to be a superkey — the same shape of rule as BCNF, one level stronger, so 4NF always implies BCNF.
- EmpSkillsLangs cross-multiplies two independent facts into spurious rows; splitting into EmpSkills and EmpLanguages fixes it, guaranteed lossless by Fagin's theorem.
- 4NF decomposition removes physical redundancy and its update anomalies — it does not remove any information the joined relations can express.